9.2 58) Given some $\varepsilon > 0$, choose N such that $N > 1/\sqrt{\varepsilon}$.

When $n > N$, $1/n^2 < 1/N^2 < (\sqrt{\varepsilon})^2/1$. Thus $|1/n^2 - 0| < \varepsilon$.

So $\lim\limits_{n\to\infty} 1/n^2 = 0$ by the definition of a limit.

9.3 26) $\displaystyle\sum_{m=2}^{\infty} \frac{5}{2^m} = \sum_{k=0}^{\infty} 5/2^k - \sum_{k=0}^{1} 5/2^k$.

$r = 1/2 < 1$ & $a = 1$

$a\dfrac{1}{1-r} \Rightarrow \dfrac{1}{1-1/2}$

$\displaystyle = 5\sum_{k=0}^{\infty} 1/2^k - (5/2 + 5)$

$= 5(2) - 15/2$

$= 5/2$

or

$\displaystyle\sum_{m=2}^{\infty} 5/2^m = \sum_{k=0}^{\infty} \frac{5}{2^2}\left(\frac{1}{2}\right)^k$

$r = 1/2$

$a = 1$

$\displaystyle = 5/4 \sum_{k=0}^{\infty} \left(1/2\right)^k$

$= 5/4(2) = 5/2$

49) By method of partial fractions (8.4 pg 465)

$\displaystyle\frac{1}{(k+1)(k+2)} = \frac{1}{k+1} - \frac{1}{k+2}$. so $\displaystyle\sum_{k=1}^{\infty} \frac{1}{(k+1)(k+2)} = \sum_{k=1}^{\infty}\left(\frac{1}{k+1} - \frac{1}{k+2}\right)$

which is a telescoping series & $\lim\limits_{n\to\infty}\left\{\dfrac{1}{1+1} - \dfrac{1}{n+2}\right\} = \dfrac{1}{2}$

9.4 20) $\lim\limits_{n\to\infty}\left\{\dfrac{n^3}{n^3+1}\right\} = 1$ so then $\displaystyle\sum_{k=1}^{\infty} \frac{k^3}{k^3+1}$ diverges by the Divergence Test

24) $f(x) = \dfrac{x}{\sqrt{x^2+4}}$ is continuous & positive but ~~&~~ $f'(x) = \dfrac{4}{(x^2+4)^{3/2}} \geq 0$ for $x > 1$, so we can't use the integral test.

$\lim\limits_{x\to\infty}\left\{\dfrac{x}{\sqrt{x^2+4}}\right\} = 1$ (use l'Hopital's rule)

so then $\displaystyle\sum_{k=1}^{\infty} \frac{k}{\sqrt{k^2+4}}$ diverge by the Divergence Test.